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RESEARCH ARTICLE



Telmisartan targets Nrf2-HO1 axis in MASLD modulating oxidative stress, inflammation, and mitochondrial dysfunction: mechanistic insights

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ABSTRACT

Metabolically dysfunction-associated steatotic liver disease (MASLD) has emerged as the leading chronic liver disease worldwide, driven primarily by metabolic derangement. The current investigation proposes to elucidate the hepatoprotective mechanisms of telmisartan (TEL) in MASLD. Twenty-four male Wistar rats were allocated to four groups (Control, MASLD, TEL-treated, and MASLD/TEL). Lipid profiles, glycemic markers, liver enzymes, and hepatic markers of oxidative stress (malondialdehyde (MDA), glutathione (GSH), superoxide dismutase (SOD), and catalase) were measured. MASLD-associated genes retrieved from GeneCards[®] were mapped to *Rattus norvegicus* ortholog genes using the gprofiler2 R package. Kyoto Encyclopedia of Genes and Genomes (KEGG) and Reactome enrichment analyses were subsequently performed using the clusterProfiler R package. Hepatic inflammatory cytokines (TNF- β , IL-6, and NF- κ B), mitochondrial respiratory enzymes (Complexes I–IV), and the gene expression of Nrf2, HO-1, MMP-9, and TIMP-1 were evaluated using spectrophotometric assays for mitochondrial respiratory enzymes. Histological assessment was done using Hematoxylin and Eosin (H&E) staining, Masson's trichrome, and TGF- β immunostaining. TEL ameliorates MASLD-associated disturbances in the serum ALT level and lipid profile. It significantly reduces the levels of oxidative stress markers. KEGG and Reactome enrichment highlighted pathways involved in lipid metabolism, insulin resistance, and inflammation, with the peroxisome proliferator-activated receptor (PPAR) and AMP-activated protein kinase (AMPK) signaling pathways being the most enriched. TEL treatment increased the hepatic expression of Nrf2, HO-1, and TIMP-1 while decreasing the expression of MMP-9. The levels of the proinflammatory cytokines TNF- α and IL-6 decreased. The activities of mitochondrial enzymes (citrate synthase and complex I) improved. MASLD induced marked hepatic fibrosis, which was markedly improved following TEL treatment. TEL has notable hepatoprotective properties in MASLD by enhancing metabolic parameters, decreasing oxidative stress, and moderating inflammatory reactions.

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MASLD; telmisartan; oxidative stress; TNF- α ; IL-6; hepatoprotective; inflammation

1. Introduction

Presently referred to as metabolic dysfunction-associated steatotic liver disease (MASLD), the disease has undergone considerable evolution in terminology and definition over the past couple of decades [1]. It was initially described in 1980 as nonalcoholic steatohepatitis (NASH), which includes steatosis, inflammation,

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hepatocellular ballooning, and fibrosis, and then combined with a more generalized nonalcoholic fatty liver disease (NAFLD) in order to encapsulate the full histological presentation of steatosis to steatohepatitis [2]. In 2020, the first redefinition of the disease was based on metabolic dysfunction and, under the nomenclature of metabolic dysfunction-associated fatty liver disease (MAFLD), followed by the second and current redefinition as MASLD, reflecting the disease's strong association with metabolic syndrome. By 2025, MASLD will be a significant public health challenge globally, with increasing rates of obesity, diabetes, and cardiovascular disease, which progress with various disease-associated processes, including oxidative stress, lipid accumulation, and chronic inflammation [3–6]. Importantly, MASLD now includes both metabolic-related cases and cases not linked to explicit metabolic causes, which can include infections, drugs, and environmental incidences [7].

Oxidative stress is increasingly known as a major pathological factor in the pathogenesis of MASLD and is inextricably linked to mitochondrial dysfunction [8]. Increased hepatic mitochondrial function, especially fatty acid oxidation, is a compensatory response to lipid overload, which generates excess reactive oxygen species (ROS), including, but not limited to, hydrogen peroxide (H_2O_2) and superoxide anions [9]. This persistent pro-oxidant status could overcome mitochondrial antioxidant defenses and disrupt mitochondrial oxidative phosphorylation. Together, these pro-oxidative processes cause worsening of hepatic injury [10]. Additionally, oxidative stress and nitrosative stress, the imbalance between the production and removal of ROS/reactive nitrogen species (RNS) by antioxidant systems, play significant roles in hepatocellular injury and in causing the progression from simple steatosis, known as metabolic dysfunction-associated steatosis (MASL), to the highest stage of disease development, a more inflammatory stage with metabolic dysfunction-associated steatohepatitis (MASH). Thus, oxidative stress is considered both a consequence and a prominent driver of the progression of MASLD [11].

Telmisartan (TEL) is classified as an angiotensin II receptor blocker (ARB) and exhibits pharmacological properties in comparison to other ARB medications, such as a longer half-life than all other ARBs do, which provides a greater and more consistent antihypertensive effect [12,13]. Research has shown that TEL exhibits hepatoprotective properties in models of MASLD by modulating key metabolic and inflammatory pathways. One of its major mechanisms of action is the activation of PPAR δ , which promotes glucose uptake into the liver, inhibits gluconeogenesis, improves hepatic lipid metabolism, and enhances insulin sensitivity, all of which are relevant to MASLD pathogenesis [14,15]. It has also exhibited anti-inflammatory effects that additively contribute to the reduction in hepatic injury; in addition, it has been demonstrated to enhance the Nrf2/HO-1 pathway in several animal models, including diabetic nephropathy, to decrease oxidative stress, suppress proinflammatory cytokines, and enhance antioxidant defense [16]. Nrf2/HO-1 is increasingly recognized as a key protective pathway in MASLD, regulating redox status, reducing hepatic lipid accumulation, and mitigating oxidative damage [17]. These mechanistic actions of TEL provide a foundation for understanding its value as a prospective pharmacotherapy for MASLD.

Currently, bioinformatics plays a crucial role in elucidating the molecular mechanisms underlying complex disease phenotypes, such as MASLD. By systematically gathering and analyzing large biological datasets, bioinformatics allows the identification of important genes, regulatory networks, and biological signaling pathways involved in the development and progression of disease [18]. This process enables researchers to analyze large sets of genes for functional enrichment of biological processes, such as inflammation, oxidative stress, and fibrosis, which are hallmarks of MASLD pathogenesis. The progression of MASLD occurs within a dynamic inflammatory pathway framework, where bioinformatics analyses of key molecular drivers have revealed temporal patterns. Differentially expressed gene, expressed in all stages of MASLD showed a distinctive V-shaped pattern in terms of the pattern of expression during the progression of MASLD, at which early inflammation and apoptosis pathways drove the earliest stage of MASLD (e.g. IL-17, TNF, and MAPK), while hallmark inflammatory gene sets (e.g. IL-6/JAK/STAT3 and NF- κ B) exhibited inverted V-shaped activation peaking in simple steatosis and late in the progression stages of MASLD [19].

The proinflammatory cytokines tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6) play crucial roles in the development of MASLD, acting in concert to mediate the inflammatory and fibrotic responses in hepatocytes. When TNF- α levels are elevated in the body, it initiates the action of Kupffer cells and activates the NF- κ B signaling pathway, thereby increasing oxidative stress and hepatocyte toxicity/injury and subsequently triggering fibrogenesis. In preclinical models, the inhibition of TNF- α can increase insulin resistance and inflammation and reduce steatosis [20]. TNF- α also inhibits the

action of alcohol dehydrogenase, which contributes to the increase in endogenous ethanol, a characteristic of MASLD [21]. Interleukin-6 is another proinflammatory cytokine; however, it is more closely associated with prolonged duration of chronic inflammation and has been suggested to be an indicator of systemic metabolic dysfunction. Mechanistically, IL-6 increases insulin resistance by increasing the level of suppressor of cytokine signaling 3 (SOCS-3), so the insulin signaling pathway in hepatocytes is impaired [22,23].

However, treatment with TEL indeed significantly reduces these cytokines, indicating that its anti-inflammatory properties may be mediated via PPAR- γ . PPAR- γ agonists can regulate the levels of TNF- α and IL-6 by inhibiting oxidative stress and NF- κ B signaling [24]. TEL is an ARB with partial PPAR- γ agonist activity, and studies have shown its ability to inhibit NF- κ B activation and subsequent inflammation-related gene expression [13]. Furthermore, TEL stimulates lipid metabolism and returns adipokine balance, with the eventual downstream effect of greater reductions in hepatic inflammation. Thus, inhibition of TNF- α and IL-6 by targeting them with TEL may have a therapeutic impact on MASLD by improving metabolic and inflammatory parameters [12].

Therefore, our current study aims to elucidate the most likely mechanisms of TEL's systemic and hepatoprotective actions in MASLD in a Wistar rat model by examining its non-antihypertensive effects. Additionally, due to the frequent overlap of MASLD with metabolic syndrome, we aimed to assess the overall metabolic effect of TEL. As we considered ways to broaden our understanding of this complex disease, we retrieved MASLD-associated genes. We subsequently performed functional pathway enrichment analyses, with a focus on biological pathways related to disease progression. The use of these bioinformatics tools guided the exploration of molecular mechanisms and prepared the way for building experimental work, providing a more holistic interpretation and understanding that ultimately facilitated the development of future therapeutics.

2. Materials and methods

2.1 Chemicals used

TEL was obtained from the German pharmaceutical company Boehringer Ingelheim. Cholesterol and bile salts were supplied by Sigma-Aldrich (USA). All the compounds used in the investigation were of the highest commercially available quality.

2.2 Animals used

Twenty-four male Wistar rats, each weighing 200–250 grams, were procured from the Holding Company for Biological Products and Vaccines located in Helwan, Egypt. The rats were kept in conventional settings with a 12-hour light–dark cycle and unrestricted access to standard chow and water.

2.3 Ethical approval

The research received approval from the Research Ethics Committee for Experimental and Clinical Studies at Ain Shams University's Faculty of Medicine (Approval code: FMASU R01/2023). All steps were taken in accordance with the principles of the Animals (Scientific Procedures) Act 1986 and associated guidelines, as well as the EU Directive 2010/63/EU for animal studies. All efforts were made to improve the environment of the animals and reduce their suffering.

2.4 Animal grouping

Following a one-week acclimatization period, the rats were randomly assigned to one of the following experimental groups ($n = 6$ per group).

- Control group. A controlled basal diet consisting of 20% protein, 10% fat, and 70% carbohydrates was maintained for a 12-week experimental period. The rats were given distilled water by oral gavage for the last four weeks of the study.
- TEL group. The rats were fed the basal diet as previously described for 12 weeks and were provided with the drug TEL (10 mg/kg/day) by oral gavage for the last four weeks. TEL was diluted in distilled water.
- MASLD group. A high-fat diet (HFD) was maintained for 12 weeks to produce MASLD. The HFD consisted of standard chow, lard, cholesterol, and bile salts (**72.8%**, 25%, 2%, and 0.2%, respectively). For the last four weeks, the animals were given distilled water by oral gavage to mimic the TEL group [16,25,26].
- MASLD/TEL group. The rats were fed a HFD for 12 weeks, while TEL (10 mg/kg/daily) was given via oral gavage for the last four weeks.

Body weight was recorded at the beginning of the study and then weekly to monitor changes accurately and adjust the TEL dose prior to and at the end of the 12-week period at sacrifice.

2.5 Sample collection

After fasting for 12 hours, the rats were anesthetized using an intraperitoneal injection of urethane (1.2 g/kg, body weight) and euthanized. Blood was obtained from the retro-orbital plexus and centrifuged for 15 minutes at 3000 rpm (approximately $1500 \times g$) to separate the plasma, which was then stored for further analysis. The liver was removed, washed in saline to remove any excess blood, blotted dry to remove any liquid, and weighed. Segments of the liver were utilized for biochemical studies, while additional portions were kept in 10% buffered formalin for histopathological examination [27]. The liver index was calculated as follows: $\text{liver index (\%)} = (\text{liver weight (g)} / \text{body weight (g)}) \times 100$.

2.6 Determination of lipid profile and glycemic parameters

Total cholesterol (TC), triglyceride (TG), high-density lipoprotein cholesterol (HDL-C), and fasting blood glucose (FBG) levels were measured in plasma samples using commercially available colorimetric kits and analyzed with a Beckman Synchron CX5 autoanalyzer (Beckman Coulter, USA). The low-density lipoprotein cholesterol (LDL-C) levels were calculated using the Friedewald equation: $LDL-C = TC - HDL-C - (TG/5)$.

The fasting blood insulin (FBI) levels were determined using an ELISA kit specific for rat insulin (Cat. No. CEA448Ra, Cloud-Clone Corp., Houston, TX, USA) (Friedewald, Levy, and Fredrickson 1972).

2.7 Determination of plasma liver transaminases

Plasma liver transaminases, alanine transaminase (ALT), and aspartate transaminase (AST) were assessed using commercial colorimetric kits. Quantification was performed on a Beckman Synchron CX5 autoanalyzer (Beckman Coulter, USA) according to the manufacturer's instructions. The assays provided quantification for assessing hepatic injury, which was expressed in units per liter (U/L).

2.8 Hepatic lipid peroxidation and antioxidant markers evaluation

Hepatic lipid peroxidation was assessed by measuring the level of malondialdehyde (MDA) using the OxiSelect™ TBARS Assay Kit (Cell Biolabs Inc., USA) according to the manufacturer's directions. Moreover, hepatic antioxidant enzymes, including glutathione (GSH), catalase, and superoxide dismutase (SOD) [28], were estimated using colorimetric reagent kits from Bio Diagnostic Company, Egypt. All assays were executed as directed by the provided studies, and absorbance was measured using a spectrophotometer.

2.9 Bioinformatics analysis

2.9.1 Retrieval of MASLD-Associated Genes

To identify genes involved in MASLD, the GeneCards human gene database (<https://www.genecards.org>), which combines information from various databases, including genomic, transcriptomic, proteomic, and functional annotations, was utilized [29]. We retrieved MASLD-related genes. Then, as we focused only on protein-coding genes to limit complexity and ensure functional relevance, we filtered gene annotations to retain only protein-coding gene entries, creating the final curated list of genes associated with MASLD for bioinformatics analysis.

2.9.2 Gene Mapping

To map the human gene list to *Rattus norvegicus* genes, human gene identifiers were mapped to rat orthologs using the gprofiler2 package in R software [30]. Initially, the gene list was curated to ensure that all human gene identifiers were valid and suitable for conversion. The gprofiler2 tool was employed to perform the conversion and map human genes to their corresponding rat orthologs, ensuring that the correct species was targeted.

2.9.3 Functional Enrichment Analysis

This was performed to detect biological pathways significantly enriched with respect to the input gene list using the clusterProfiler package in R [31] for Kyoto Encyclopedia of Genes and Genomes (KEGG) and Reactome using the annotation database specific for *Rattus norvegicus*, with a *p* value cutoff of 0.05 and a Benjamini–Hochberg adjusted *q* value cutoff of 0.05 for correction of multiple testing. The results were visualized using R software via bar plots and a gene-concept network view, which represented the genes and enriched pathways.

2.10 Determination of hepatic inflammatory markers

Liver levels of TNF- β , IL-6, and NF- κ B were determined using ELISA kits (MyBioSource, USA). The liver tissues were homogenized and then extracted according to protocols generated by the manufacturer. The concentrations of the three inflammatory biomarkers were determined by taking the absorbance values obtained and comparing them to a standard curve constructed. All the samples were assessed in duplicate to ensure that all the collected data were accurate and reproducible [32].

2.11 Determination of the activity of hepatic mitochondrial respiratory chain enzymes

Hepatic mitochondria were isolated via a modified Krause method (Krause et al., 2005). Liver tissue was homogenized in 250 mM sucrose, 50 mM Tris-HCl, 5 mM EDTA, and 0.5 mM PMSF buffer (pH 7.4) and centrifuged at $800 \times g$ for 10 min at 4 °C. The supernatant was buffered again and centrifuged at $8000 \times g$ for 10 min at 4 °C. The mitochondrial pellets were washed and subsequently resuspended in 250 mM sucrose with 0.5 mM PMSF. The protein concentration of the mitochondrial preparation was assessed via the Lowry method [33]. The hepatic activity of complex I (NADH dehydrogenase) was assessed by measuring the rate of NADH-dependent ferricyanide reduction at 420 nm [34]. The liver activity of complex II (succinate dehydrogenase) was measured as the decrease in absorbance due to the reduction in 2,6-dichloroindophenol (DCIP) at 600 nm. Fischer et al. measured the liver activity of complex II-III (succinate: cytochrome C oxidoreductase) by measuring the percent reduction in cytochrome C when succinate was used as a substrate at 550 nm [35]. The hepatic activity of complex IV (cytochrome c oxidase) was measured at 550 nm by absorption decline caused by the oxidation of reduced cytochrome c, which was produced by reducing cytochrome with NaBH₄ and HCl [36].

2.12 *Nrf2, HO-1, MMP-9, and TIMP-1 mRNA quantification by real-time reverse transcription-PCR (qRT-PCR)*

The extraction of total RNA was performed using the QIAzol reagent following the manufacturer's instructions (Qiagen, Germany). The concentration and purity of the RNA were measured using the Thermo Scientific NanoDrop 2000 (USA). The purified RNA was converted to complementary DNA (cDNA) using the Bioline cDNA synthesis kit (Bioline, USA). The polymerase chain reaction (PCR) protocol consisted of the following steps: 2 minutes at 95 °C, followed by 40 cycles of denaturation (10 seconds at 95 °C) and annealing and extension combined (30 seconds at 60 °C). The primer sequences used in this study are shown in Table 1. The primers used were purchased from Vivantis Technologies (Malaysia). The expression level of miRNA was normalized against that of β -actin and calculated using the $2^{-\Delta\Delta CT}$ method.

2.13 *Histopathological Examination: preparation of paraffin blocks and basic stains*

Liver tissues were obtained and fixed in 10% neutral buffered formalin. The tissues were dehydrated through a series of alcohol steps and then cleared in xylene. The tissues were embedded in paraffin and then cut into 4- μ m-thick slices using a microtome, resulting in tissue blocks. The tissue sections were mounted on glass slides and stained with hematoxylin and eosin, and the collagen fibers were stained with Masson's trichrome. Immunohistochemical stains were used to examine protein expression [40].

2.14 *Immunohistochemistry analysis of fibrotic marker TGF- β*

The streptavidin–biotin method with peroxidase labeling was used for immunohistochemical studies. The slices were blocked in 1.5% normal goat serum in PBS. For the detection of tissue fibrosis, an anti-TGF- β rabbit monoclonal antibody (clone TB21, MCA 797, Serotec, Oxford, UK, as listed by the manufacturer at a 1:200 dilution) was used. Before incubation with a biotinylated secondary antibody, all the slides were thoroughly washed with PBS. The substrate chromogen used was 3,3'-diaminobenzidine (DAB), for which a brown staining reaction (indicating positive staining) could typically be observed in the cytoplasm and rarely in the nucleus. All the slides were counterstained with hematoxylin, dried, and mounted [41].

2.15 *Morphometric evaluation of Masson trichrome and immunohistochemically stained sections*

For each rat in each group, five randomly chosen sections were analyzed using a 40X objective and a 0.071 mm² area of analysis. We also took into account the homogeneity of section thickness to allow sufficient and viable comparisons. The images of the sections were taken with a digital camera (Toup Cam type BX53; Japan) connected to a computer. The light microscopic camera used was an Olympus BX53 (Tokyo, Japan). ImageJ v. 1.48 was used to analyze the collagen tissue in the sections stained with Masson's trichrome and immunologically positive reactions to α -SMA and TGF- β [42].

Table 1. Primers sequences used in RT-PCR studies.

Gene	Sequence	References
Nrf2	5'-TTGTAGATGACCATGAGTCGC-3' 5'-TGCCTGCTGTATGCTGCTT-3	[37]
HO-1	5'-GTAAATGCAGTGTGGCCCC-3' 5'-ATGTGCCAGGCATCTCCTTC-3	[37]
MMP-9	5'-TCGAAGGCGACCTCAAGTG 3' 5'-TTCGGTGTAGCTT TGGATCCA-3	[38]
TIMP-1	5'-CTGAGAAGGGCTACCAGAGC-3' 5'-TATTGCCAGGTGCACAAATC-3.	[39]
β -actin	5'-TGGGAAGCTGGTCATCAAC-3' 5'-CTCCAGTGTGGTGAA-3	[39]

2.16 Statistical analysis

The data are presented as the means and standard errors of the mean (SEM) and were analyzed using one-way ANOVA followed by post hoc Tukey's test for comparison. A significance level of ($P < 0.05$) was defined as statistically significant. The analysis was performed using GraphPad Prism 9.0.0 (121) software program for Windows, GraphPad Software, Boston, Massachusetts, USA.

3. Results

3.1 Effect of different experiments on hepatic index, glycemic control, and lipid profile in HFD-induced MASLD in Wistar Rats

To evaluate the effects of different experiments on various hepatic functions, the hepatic indices, GCs, and lipid profiles were assessed (Figure 1). TEL administration significantly improved the metabolic and hepatic measures associated with a HFD in the MASLD group. Compared to control rats, the MASLD group significantly increased liver weight ratio (4.05 ± 0.34 vs. $2.57 \pm 0.21\%$), serum glucose (293.46 ± 15.26 vs. 71.58 ± 3.52 mg/dL), insulin levels (39.07 ± 5.38 vs. 2.56 ± 0.63 μ U/mL), triglycerides (101.05 ± 11.25 vs. 51.48 ± 2.57 mg/dL), total cholesterol (115.51 ± 6.41 vs. 64.61 ± 2.71 mg/dL) and LDL cholesterol (78.19 ± 7.22 vs. 21.19 ± 1.08 mg/dL) with significant decrease in HDL levels (19.37 ± 2.55 vs. 34.21 ± 1.50 mg/dL) ($p < 0.05$ for all). In the TEL treatment in MASLD rats significantly revoked these changes including liver weight ratio ($2.71 \pm 0.19\%$), serum glucose (150.63 ± 8.54 mg/dL), insulin (21.62 ± 1.49 μ U/mL), triglycerides (87.71 ± 8.14 mg/dL), total cholesterol (90.50 ± 5.92 mg/dL) and LDL (59.52 ± 5.39 mg/dL) while increasing HDL to (25.66 ± 3.61 mg/dL) ($p < 0.05$ vs. MASLD group). Data from this study suggest a

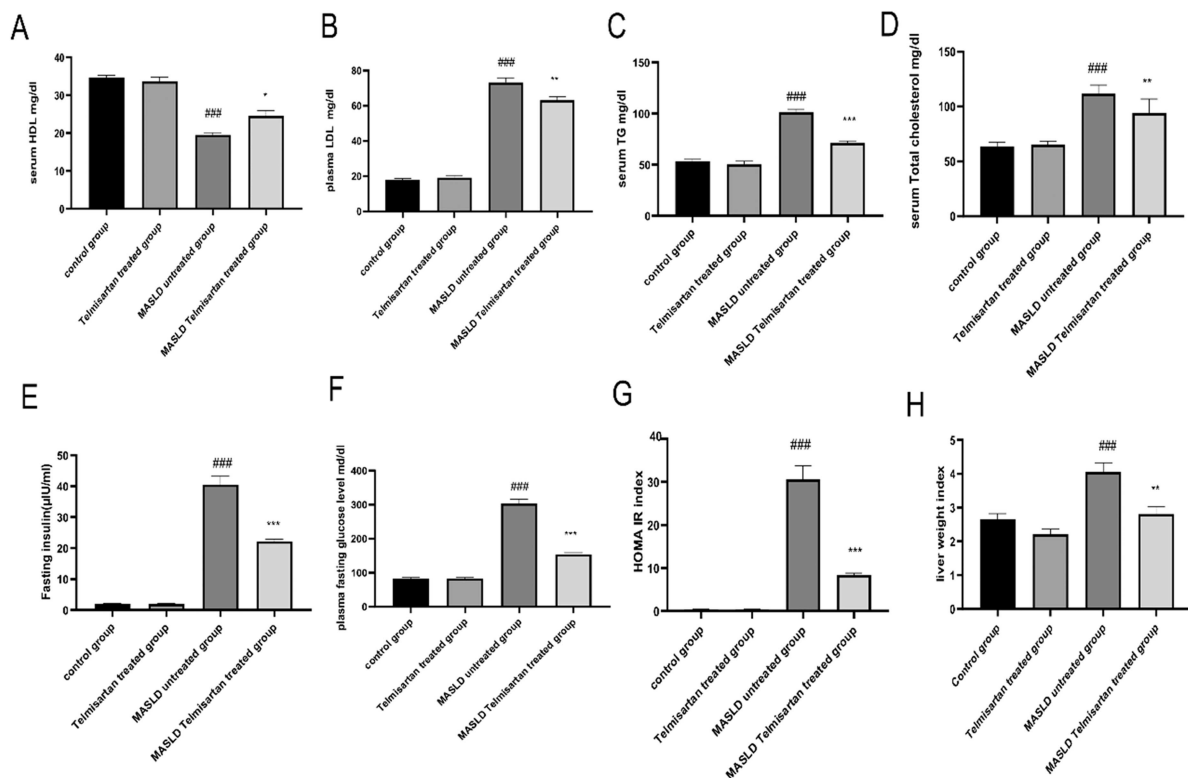


Figure 1. Effect of different experiments on (A) High-Density Lipoprotein (HDL) levels, (B) Low-Density Lipoprotein (LDL), (C) triglycerides (TG), (D) total cholesterol, (E) fasting insulin, (F) fasting glucose, (G) Homeostatic Model Assessment for Insulin Resistance (HOMA IR), and (H) liver weight ratio in High-Fat Diet (HFD) induced Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD) rats. The data are presented as the mean $\pm$ SEM. Statistical significance was determined by one-way ANOVA followed by Tukey's post hoc test. # $p < 0.05$ vs. Control; * $p < 0.05$ vs. MASLD.

protective effect of TEL with respect to dyslipidemia, insulin resistance, and liver enlargement in MASLD following high-fat diet administration.

3.2 Effect of telmisartan on liver enzymes (ALT and AST) in MASLD Wistar rats

Investigating the effects of TEL on liver enzymes were evaluated in this study. The levels of ALT and AST in the serum were significantly higher in the MASLD-untreated group, with an average ALT concentration of 143.83 U/L and an average AST concentration of 237.5 U/L, than in the control groups, which presented average ALT and AST levels of 16–17 U/L and 17.83 U/L, respectively (Figure 2). This elevation suggests significant hepatocellular injury related to MASLD. In addition, a significant reduction in TEL in the MASLD experiment was observed for liver enzymes, where ALT decreased to approximately 92.17 U/L; however, ALT was still significantly higher than the ALT control levels, and AST decreased to 161.83 U/L, indicating the hepatoprotective effect of telmisartan. Furthermore, when TEL was administered to healthy rats, ALT or AST levels did not increase, indicating that TEL is safe in a nondiseased state. Collectively, these findings support a potential therapeutic role for TEL in improving liver damage associated with MASLD.

3.3 Effect of telmisartan on oxidative stress markers in MASLD rats

To evaluate hepatic oxidative damage, we assessed oxidative stress markers (Figure 3). Biochemical evaluation of oxidative stress markers revealed a pronounced imbalance in the MASLD-untreated group compared to the control group. The MDA levels were significantly greater in the MASLD experimental group (40.9–61.2 $\mu\text{mol/L}$) compared to the control group (10.4–15.2 $\mu\text{mol/L}$). TEL treatment reduced the MDA level to 27–33 $\mu\text{mol/L}$, indicating a decrease in lipid peroxidation. CAT activity was highly diminished in the MASLD group (0.01–0.04 U/mg protein) but markedly increased in the MASLD/TEL group (0.13–0.19 U/mg), although it was still lower than that in the control group (~0.20–0.23 U/mg). The concentrations of GSH, a second measure of antioxidant activity, were also depleted in the MASLD group (4.8–5.8 $\mu\text{mol/g}$ tissue) compared to controls (~10 $\mu\text{mol/g}$) but were partially restored with TEL treatment (5.7–8.2 $\mu\text{mol/g}$). SOD activity decreased in MASLD (9.6–10.5 U/mg protein) compared with the control (13.4–16.2 U/mg) but increased with TEL treatment (11.7–13.9 U/mg). As a result, our data indicated that TEL may reduce oxidative stress in MASLD by increasing antioxidant capacity and lowering lipid peroxidation.

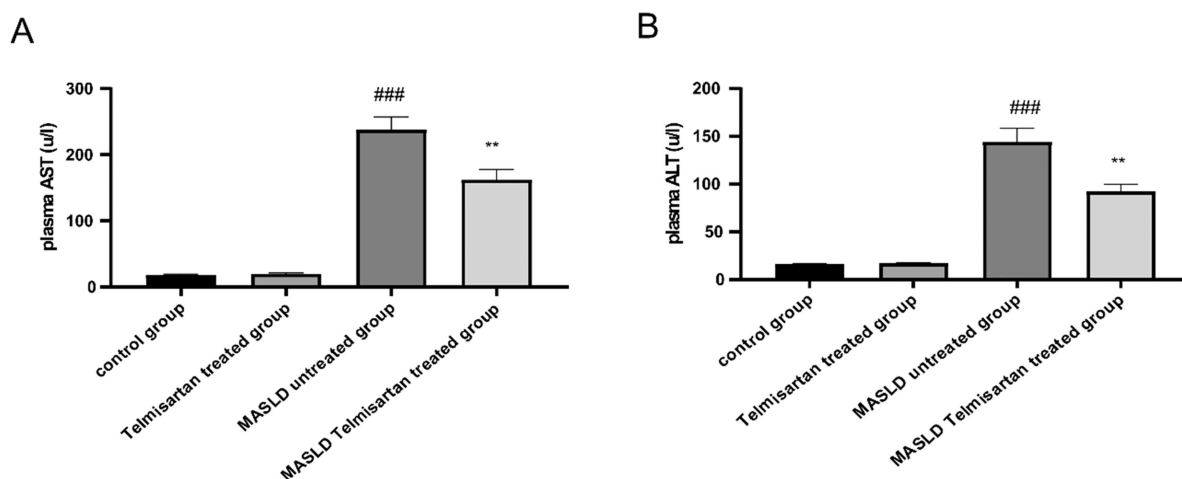


Figure 2. Effect of telmisartan (TEL) on liver enzymes in high-fat diet (HFD)-induced metabolic dysfunction-associated steatotic liver disease (MASLD). (A) Plasma aspartate transaminase (AST) and (B) plasma alanine aminotransferase (ALT) levels. #Significance relative to the control and *significance relative to MASLD according to one-way ANOVA followed by post hoc Tukey's test. The data are presented as the means and standard errors of the means (SEMs).

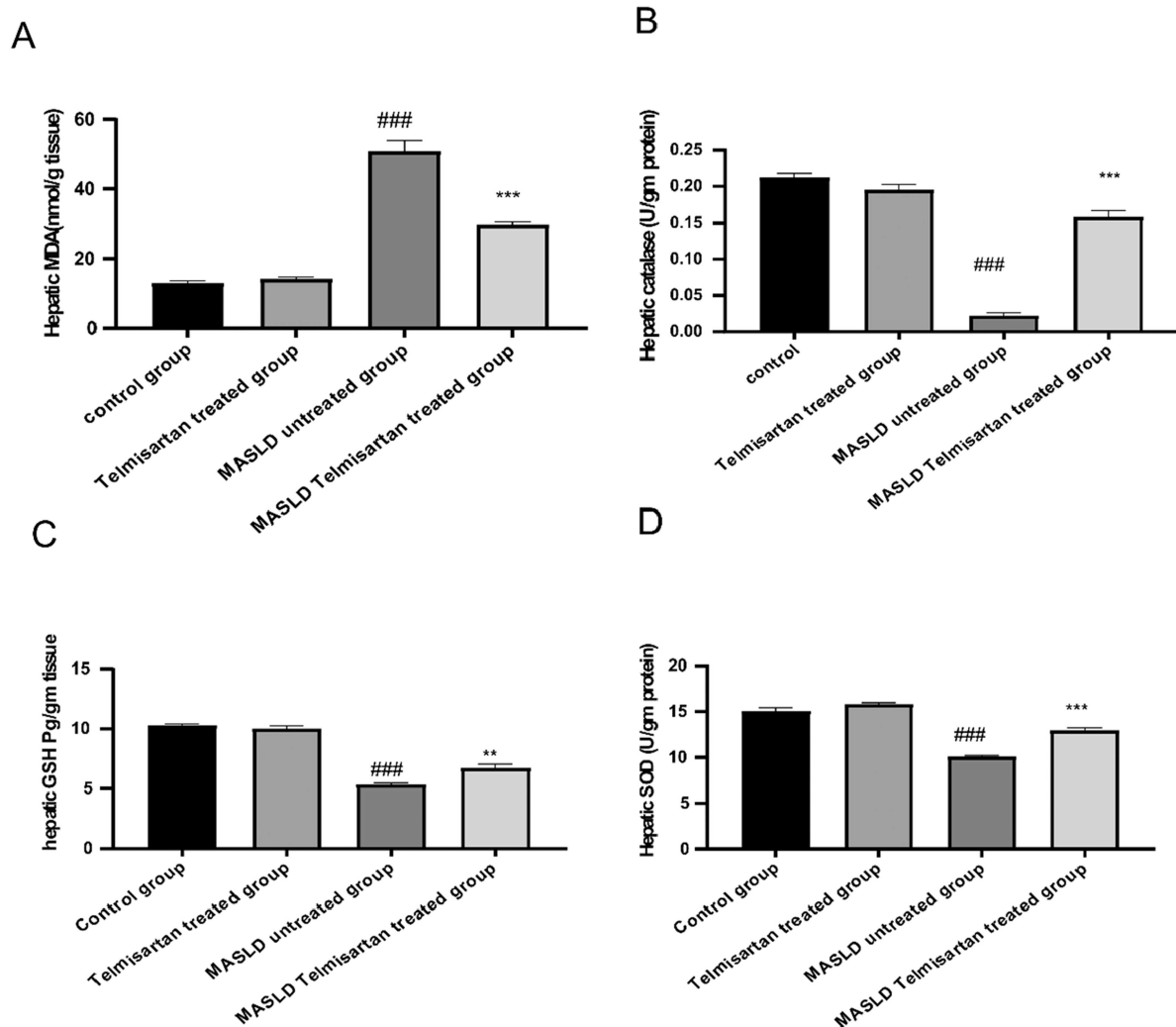


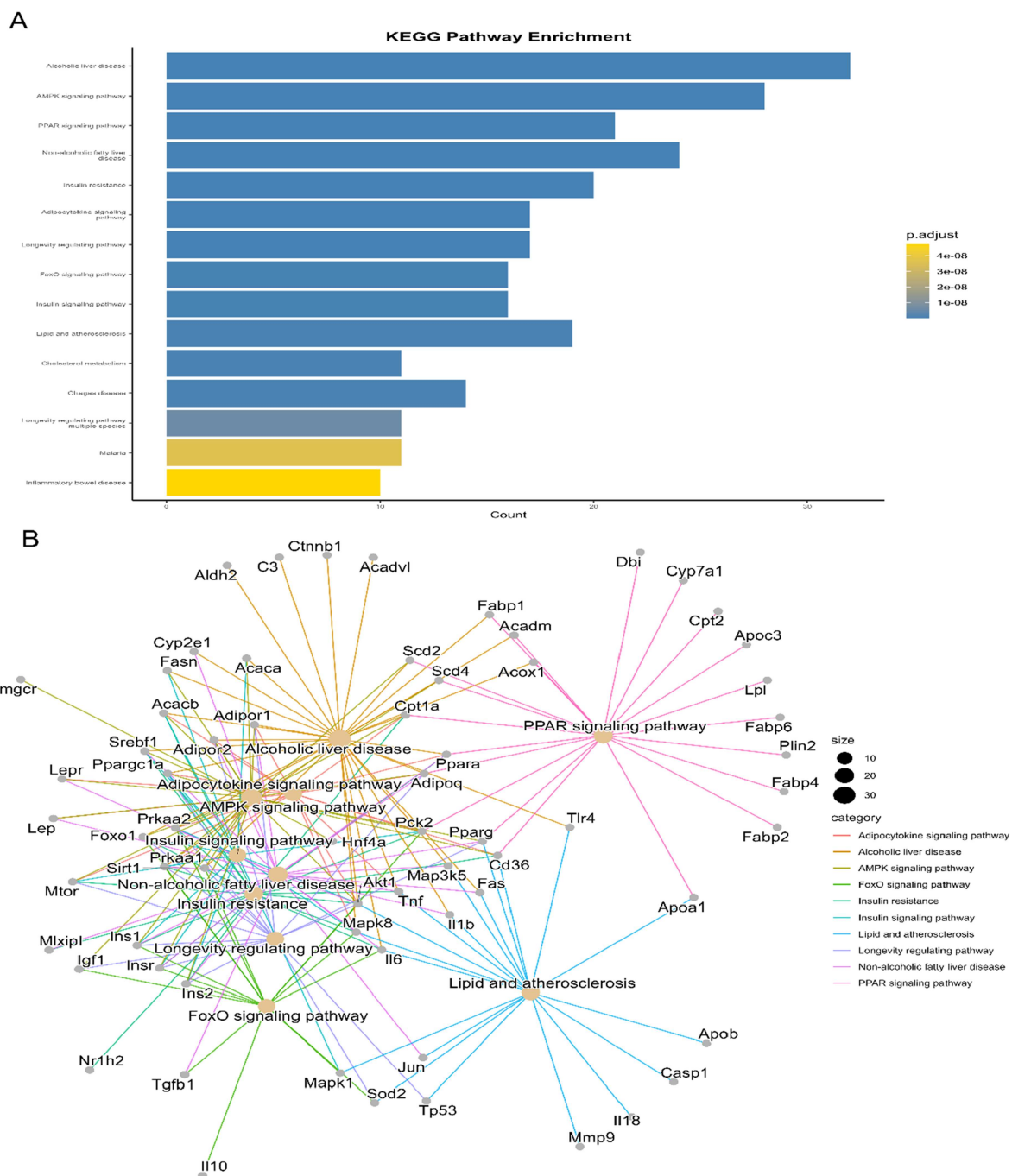
Figure 3. Effect of telmisartan (TEL) on hepatic oxidant and antioxidant status in high-fat diet (HFD)-induced metabolic dysfunction-associated steatotic liver disease (MASLD) patients. Hepatic malondialdehyde (MDA), catalase, glutathione (GSH), and superoxide dismutase (SOD) are shown (A–D), respectively. #Significance against control and *significance against NAFLD according to one-way ANOVA followed by post hoc Tukey's test. The data are expressed as the means and SEMs.

3.4 Bioinformatics analysis

To better understand the molecular mechanisms underlying MASLD pathogenesis, our study focused specifically on protein-coding genes. Protein-coding genes are those that encode functional proteins that act and carry out relevant biological processes, such as inflammation, lipid metabolism, oxidative stress, and fibrosis, which ultimately drive MASLD. This focus will be useful for identifying verifiable information from the bioinformatics route to experimental work and clinical relevance.

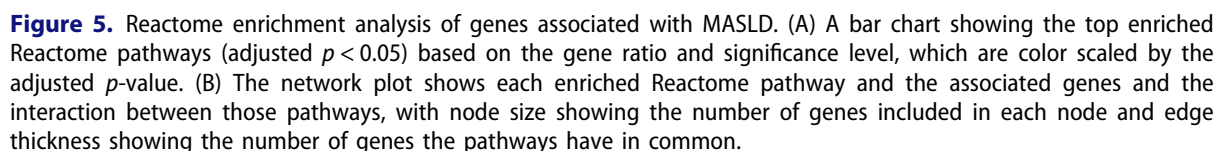
A total of 218 MASLD-associated genes were identified from the GeneCards human disease database. As we focused only on protein-coding genes to limit complexity and ensure functional relevance, a final list of 153 protein-coding genes was used for further analyses. The rat orthologs underwent functional enrichment analysis using KEGG and Reactome pathways. The enrichment analysis emphasized the functional relevance of these genes in the molecular mechanism behind MASLD in the rat model.

KEGG pathway enrichment analysis revealed that the analyzed gene set was significantly overrepresented in pathways related to inflammation and metabolism (Figure 4). Pathways that were highly enriched included alcoholic liver disease and nonalcoholic fatty liver disease. In addition to AMPK signaling and PPAR signaling, insulin resistance was the most enriched pathway, with highly significant adjusted *p* values



(FDR < 0.01). In particular, multiple genes of interest from the context of MASLD were located within these pathways, including Il6, TNF, Il1b, Mapk8, and Map3k5, which are recognized mediators of hepatic inflammation and metabolic dysregulation. Although Nrf2, HO-1, Mmp9, and Timp1 were not directly listed, the presence of their upstream regulators (i.e. Mapk8, TNF) indicates that their roles in activated inflammatory cascades are likely (Figure 5).

The Reactome pathway enrichment analysis identified thirty-four enriched pathways (adjusted $p < 0.05$) among the differentially expressed protein-coding genes. The most significant difference was the



metabolism of lipids, which encompassed thirty-five out of the one hundred and three genes included. These findings indicate that lipid dysregulation is the main aspect of MASLD pathogenesis. In addition, significant pathways related to lipids, including fatty acid metabolism, triglyceride metabolism, and carnitine metabolism, were also observed, each of which is related to lipid handling in the liver and energy homeostasis. Pnpla3, Cpt1a, Acadm, Ppara, Srebf1, and Fasn were identified as important metabolic regulators that were shared hits across multiple pathways. The results indicate that the tested genes related to MASLD are largely involved in lipid biosynthesis, β -oxidation, and lipid transport, providing a mechanistic basis for hepatic steatosis and its progression.

The number of pathways enriched in our study highlights the potential for studying the role of these protein-coding genes to better understand the molecular changes associated with this complex disease and identify potential therapeutic candidates. Collectively, these results demonstrate the biological significance of these selected genes and reaffirm their potential contributions to MASLD disease progression, which can be used for future molecular validation experiments.

3.5 Effect of TEL on hepatic inflammatory markers in HFD-induced MASLD in Wistar rats

To evaluate the effects of the most important proinflammatory cytokines related to the disease state, TNF- α and IL-6, which are key contributors as part of their synergistic roles in MASLD pathogenesis, were evaluated using different experimental treatments (Figure 6). The MASLD experimental group presented significantly increased levels of TNF- α and IL-6 in serum compared to the control one, indicating that a strong inflammatory response is associated with the disease state. The serum levels of TNF- α increased from an average value of 25.1 pg/mL in the control group to 43.5 pg/mL in the MASLD group. With TEL treatment in MASLD rats, we observed a decrease in TNF- α levels of 30.6 pg/mL. Although the TNF- α levels in the MASLD and MASLD-treated groups were still elevated compared to controls, this nonetheless decreased. Furthermore, the serum IL-6 levels in the MASLD experimental group were significantly higher (101.6 pg/mL) than those in the control group (40.4 pg/mL), with treatment reducing the IL-6 concentration to 58.1 pg/mL, indicating an anti-inflammatory effect. TEL without MASLD induction did not significantly affect cytokines in healthy rats, whereby its modulatory effect was limited to a disease state or condition.

3.6 Effect of TEL on hepatic Nrf2, HO-1, MMP-9, and TIMP-1 mRNA gene expressions in HFD-induced MASLD in Wistar rats

To determine the molecular effect of TEL on oxidative stress and extracellular matrix remodeling, we examined the hepatic mRNA expression of different genes, such as Nrf2, HO-1, MMP-9, and TIMP-1, which play critical roles in antioxidant defense as well as fibrosis regulation and are severely altered in MASLD (Figure 7). Several markers of the oxidative stress response and tissue remodeling showed notable dysregulation in relation to the MASLD condition. NRF2 was significantly lower in the untreated

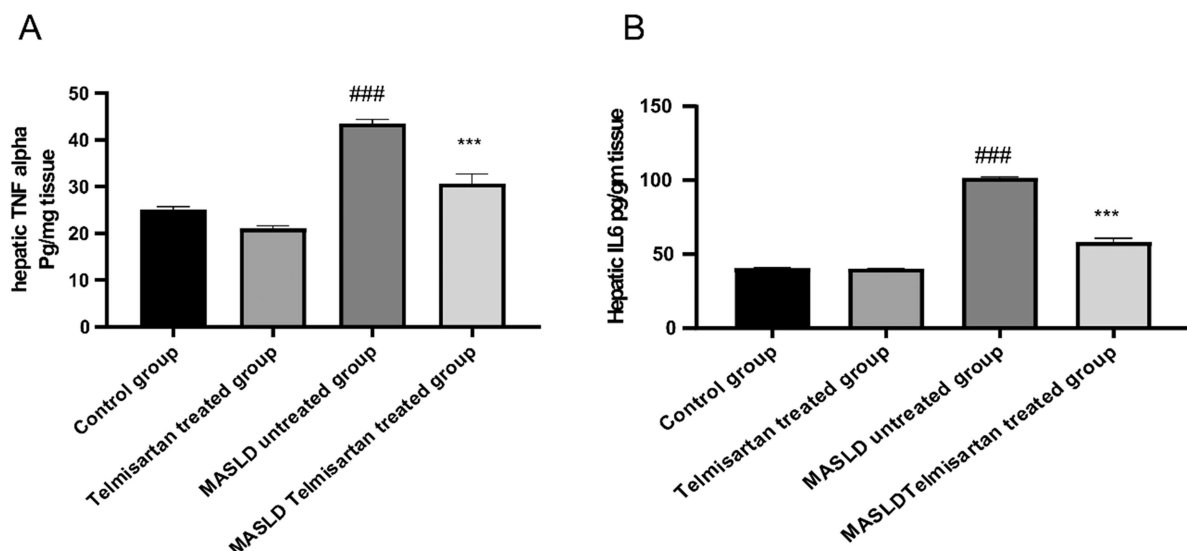


Figure 6. Effect of telmisartan (TEL) on hepatic inflammatory marker TNF- α (A) and IL-6 (B) status in high-fat diet (HFD)-induced metabolic dysfunction-associated steatotic liver disease (MASLD). #Significance against control and *significance against MASLD according to one-way ANOVA followed by post hoc Tukey's test. The data are expressed as the mean and SEM.

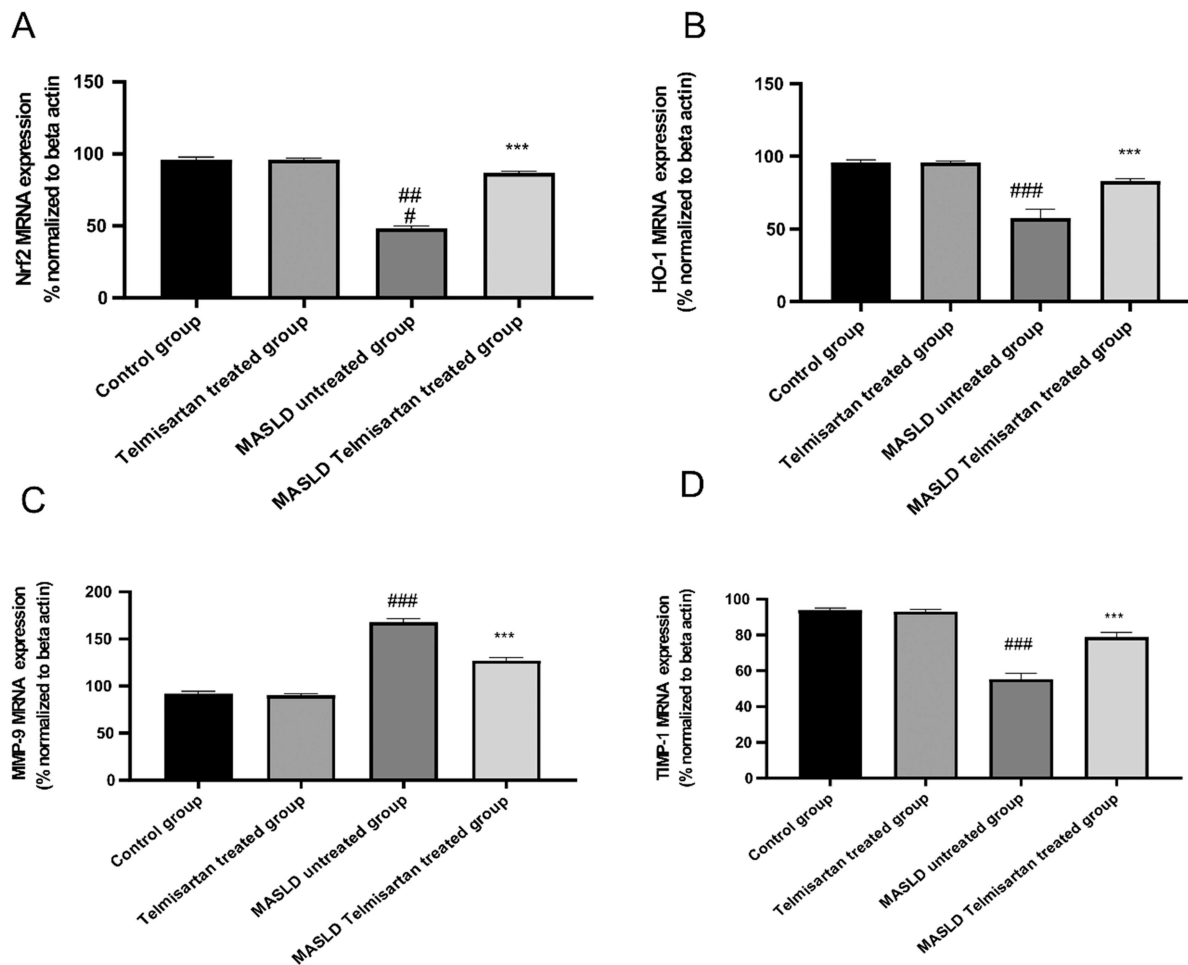


Figure 7. Effect of telmisartan (TEL) on the hepatic mRNA expression of Nrf2, HO-1, MMP-9, and TIMP-1 in high-fat diet (HFD)-induced metabolic dysfunction-associated steatotic liver disease (MASLD) (A-D). #Significance against control and *significance against MASLD according to one-way ANOVA followed by post hoc Tukey's test. The data are expressed as the mean and SEM.

MASLD group (mean of 48.0) than in the control group (mean of 95.8), suggesting a possible defect in antioxidant defense. However, the NRF2 levels (mean of 86.5) were partially restored in MASLD rats treated with TEL, which approached those of the controls. A similar observation was made for HO-1, where average HO-1 expression was decreased in MASLD rats (mean of 58.0); however, HO-1 expression was restored post-treatment (mean of 82.5), demonstrating that TEL enhances antioxidant pathways. The tissue remodeling marker MMP9 was significantly higher in the MASLD experimental group (mean of 167.5) than in the control group (mean of 91.7). TEL-treated MASLD rats presented an average expression of MMP9, which was reduced to 126.7, suggesting that TEL may be able to mitigate fibrotic activity. In addition, the level of TIMP, an inhibitor of MMPs, was also reduced in the MASLD experimental group (mean of 55.2); however, TEL treatment restored TIMP-1 to levels closer to those of the control group (mean of 78.6), suggesting a possible rebalancing event that occurred during matrix remodeling. Our results suggest that TEL may have protective effects through modulating both oxidative stress and fibrotic pathways in MASLD.

3.7 Effect of TEL on the activity of hepatic mitochondrial respiratory enzyme complexes in HFD-induced MASLD in Wistar rats

Mitochondrial dysfunction is a recognized feature of MASLD progression, especially in relation to the activity of respiratory enzyme complexes. In order to study the potential protective effect of TEL, we

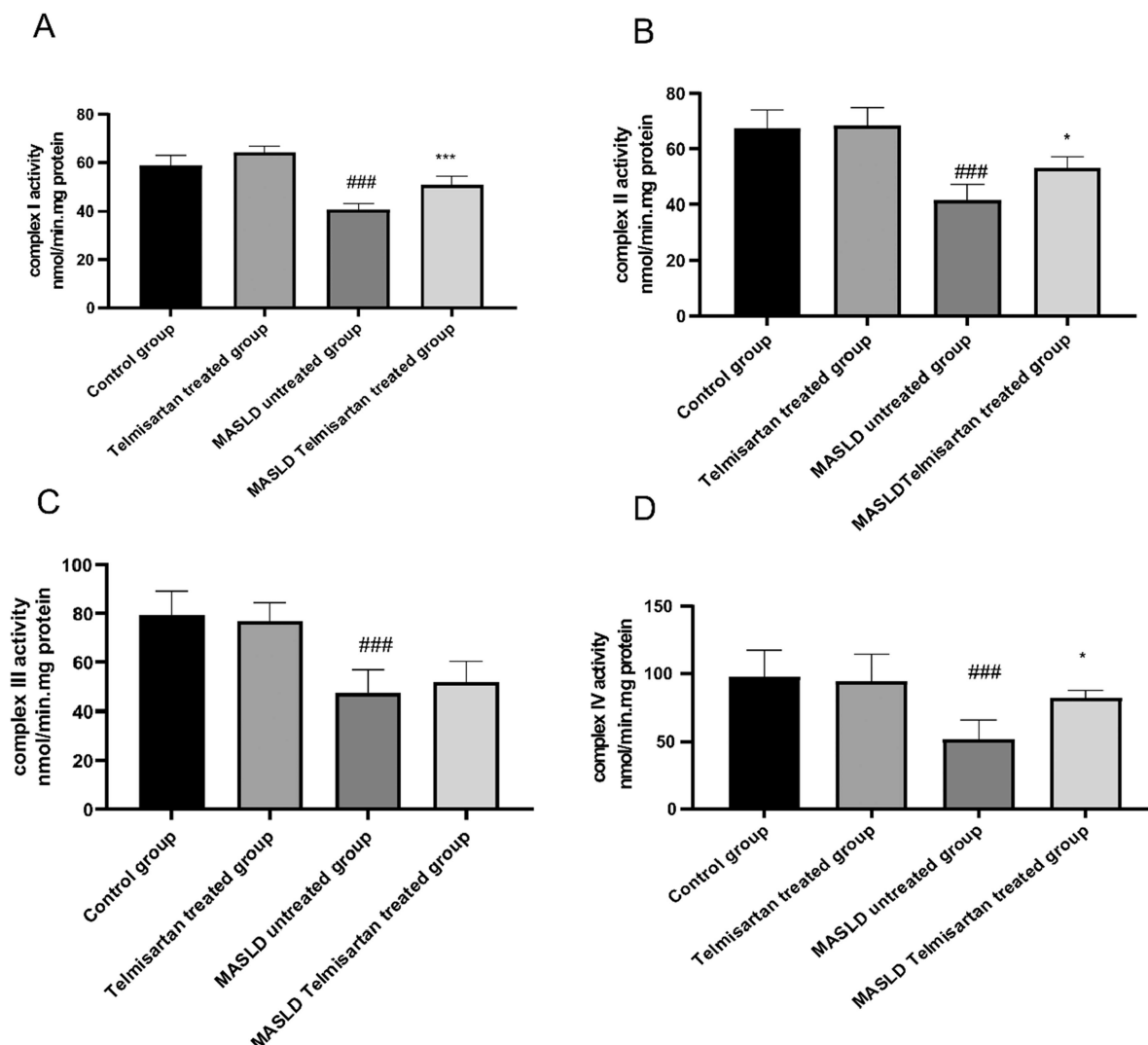


Figure 8. Effect of TEL on hepatic mitochondrial respiratory enzyme complexes in HFD-induced MASLD. #Significance against control and *significance against MASLD according to one-way ANOVA followed by post hoc Tukey's test. The data are expressed as the mean and SEM.

examined the activities of hepatic mitochondrial complexes I, II, II–III, and IV. As shown in Figure 8, a marked decrease in hepatic mitochondrial function occurred with the development of MASLD, as shown by a marked decrease in the activities of respiratory enzyme complexes I–IV. For Complex I, MASLD rats presented a mean activity of approximately 40.7, while controls presented a mean activity of approximately 58.7 ($P < 0.001$). TEL treatment significantly increased complex I activity in MASLD rats, with a mean activity of approximately 50.7 ($P < 0.001$ vs the MASLD group). A similar pattern was observed for Complex II, where MASLD rats presented a significantly decreased mean activity level of approximately 41.5 compared with that of controls (approximately 67.3). TEL administration significantly improved complex II activity to approximately 53.2 ($P < 0.05$). While Complex III mean activity was also lower in MASLD rats (47.3) compared to controls (79.2), the response in activity with TEL (51.8) was slightly less pronounced and not statistically significant in comparison to that in the MASLD group. For Complex IV, a significant mean decrease occurred in MASLD rats (51.8) compared to controls (96.2) ($P < 0.001$). The activity of Complex IV was significantly increased in TEL/MASLD rats (82.0) ($P < 0.05$), although this activity level was still significantly lower than that in control rats. Collectively, the activity analysis suggests that TEL enhances mitochondrial respiration, specifically for Complexes I, II, and IV, for livers that have a negative adaptation to MASLD.

3.8 Histopathological results

Light microscopic examination using H&E staining with different magnifications of the control and TEL-treated groups revealed normal liver microarchitecture in the form of arranged hepatocytes together with Kupffer's cells around the central vein (CV) with irregular blood sinusoids in between forming hepatic lobules. These hepatic lobules are further arranged around the portal area, which contains branches of the hepatic artery, bile duct and tributary of the portal vein. Compared to the examination of the MASLD untreated group, which showed loss of hepatic architecture and separated hepatic cords with extensive intracytoplasmic vacuolations with a signet ring appearance and dilated irregular blood sinusoids between the hepatocytes and degenerative changes in the form of apoptotic-like lesions, pyknosis, karyorrhexis, and karyolysis. On the other hand, examination of the MASLD/TEL experimental group revealed a pattern similar to that of the control and TEL-treated groups, characterized by normal hepatic architecture with minimal cytoplasmic vacuolation (Figure 9). Our findings, which are consistent with the H&E findings, suggest that TEL has a hepatoprotective effect, reducing the decline in hepatic tissues associated with the progression of MASLD disease.

Examination of Masson's trichrome-stained sections in both the control and TEL-treated groups revealed little collagen distribution in the CV and portal regions. The MASLD-untreated group showed an aggregation of collagen fibers surrounding the central vein, portal region, and hepatocytes. Conversely, MASLD/TEL revealed a limited presence of collagen fibers in the CV and portal regions. Compared with the control experimental group, the image analysis of the Masson trichrome-stained sections revealed a significant increase in the area of fibrosis in the MASLD group. When compared to the MASLD experimental group, treatment with TEL resulted in a significant decrease in the area of fibrosis. However, the fibrotic area was still slightly higher than the control area (Figure 10). The administration of TEL has antifibrotic effects. However, the inability of the fibrotic area to revert to control levels indicates that TEL reduces but does not completely return the progression of fibrosis in MASLD to control levels. Overall, TEL appears to have promising therapeutic potential for reducing fibrosis progression in MASLD.

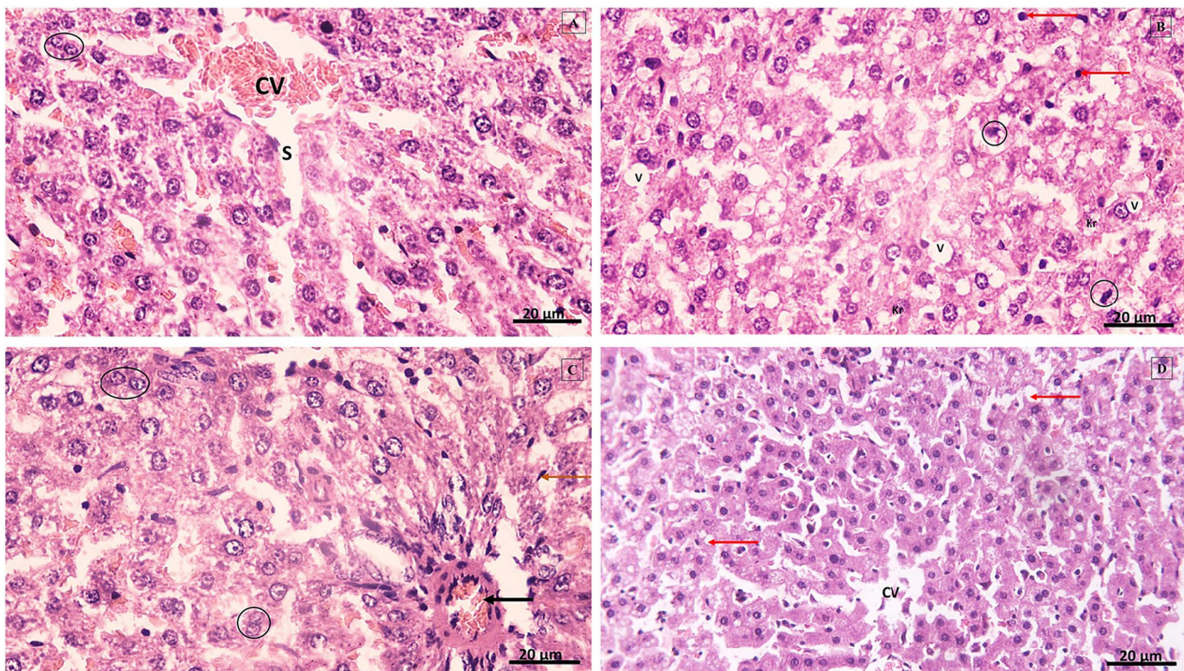


Figure 9. Histological examination of the effect of TEL on HFD-induced MASLD in different groups. A: control group, B: TEL group, showing normal hepatic architecture. Central vein (CV), blood sinusoids (S), Kupffer cells (red arrow), binucleated hepatocytes (circles), and vacuolations (V). C: the MASLD group showed dilated irregular hepatic sinusoids (S), vacuolations (V), pyknosis (red arrow), karyorrhexis (kr), and apoptotic lesions (circle). D: MASLD + TEL group, showed dilated central vein (CV) and vacuolations (red arrows).

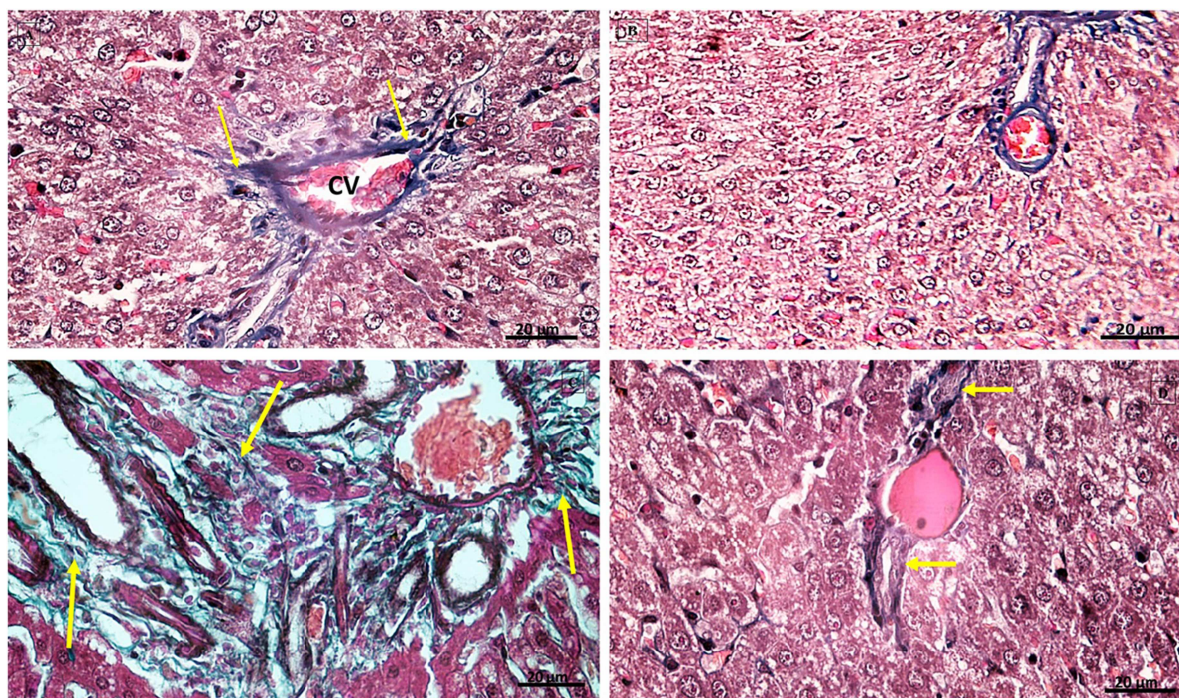


Figure 10. Masson's trichrome-stained sections show the effect of telmisartan on collagen deposition in HFD-induced MASLD in different groups. A: control group, B: TEL group, showing a low amount of collagen fibers (arrow) around the central vein (CV). C: The MASLD group has an increased amount of collagen fibers (yellow arrow), and D: the MASLD + TEL group shows a low amount of collagen fibers (yellow arrow).

The immunohistochemical study of the control and TEL-treated groups revealed a weak positive reaction for TGF- β . The untreated MASLD group showed a strong positive immunoreaction to TGF- β , while the MASLD/TEL group showed a weak positive immunoreaction to TGF- β . The mean area percentage of TGF- β in the MASLD-untreated experimental group was significantly greater than that in the control and TEL-treated groups; however, the MASLD/TEL group showed a significant decrease compared with the MASLD group. In agreement with our biochemical analysis, these results support the hepatoprotective potential of TEL in MASLD progression (Figure 11).

4. Discussion

MASLD is currently a well-known chronic liver disease characterized by excess fat in the liver. It is linked with obesity, type 2 diabetes, and cardiovascular disease. Over time, it has emerged as the most common cause of chronic liver disease as well as the common cause of liver-related morbidity and mortality [43]. Our current study investigated the hepatoprotective effects of TEL using a Wistar rat model of MASLD, thereby emphasizing its non-antihypertensive effects. TEL has not only historically been categorized as an ARB but also been found to have partial PPAR γ agonist properties, which suggests advantages not mediated by blood pressure. Our study utilized bioinformatics-led pathway enrichment, in conjunction with experimental data, to examine how TEL affects lipid metabolism, inflammation, and fibrogenesis pathways at the molecular level. We aimed to determine whether TEL has therapeutic activity in MASLD, not only as a cardiovascular agent but also to demonstrate its ability to act as a multifunctional metabolic modulator.

The progression of MASLD reflects a complex interplay between metabolic dysregulation, lipid accumulation, and oxidative stress, which are often initiated by dietary and lifestyle factors. Therefore, HFD-induced animal models are valid tools for exploring disease progression and therapeutic possibilities. With this in mind, this study aimed to examine whether TEL, a partial PPAR- γ agonist, could protect against the pathological characteristics of MASLD in Wistar rats by examining hepatic lipid accumulation, metabolic dysregulation, and oxidative stress. In the present study, induction of MASLD in experimental models, such

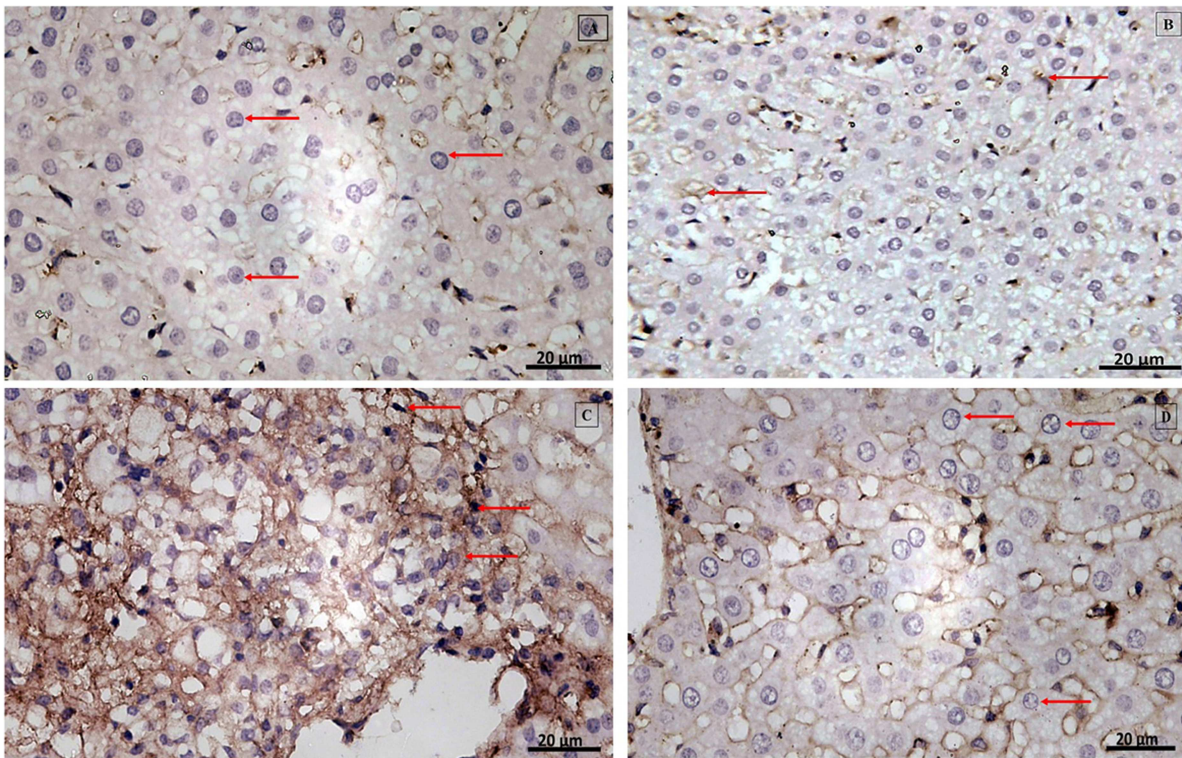


Figure 11. Immunohistochemically stained sections show the effect of TEL on the expression of TGF- β in HFD-induced MASLD. (A) Control group, (B) TEL group, negative immunoreaction, (C) MASLD group, strong positive immunoreaction, and (D) MASLD + TEL group, weak positive immunoreaction.

as HFD-fed rats, resulted in a significant increase in the liver weight index compared to controls ($P < 0.001$), reflecting hepatic steatosis and fat accumulation. TEL treatment significantly reduced the liver weight index in MASLD rats ($P < 0.01$), indicating a decrease in hepatic fat content and an improvement in liver pathology. The hepatoprotective effect of TEL can be explained via the effects of TEL on metabolic actions, such as PPAR- γ activation, which increases lipid metabolism and decreases hepatic steatosis [44,45]. The effect of TEL on hepatic fat was consistent with its broader metabolic actions, which led to a decrease in the amount of visceral fat and improved metabolic measures without affecting total food consumption or energy expenditure.

Alternatively, induction of MASLD in Wistar rats using an HFD causes marked metabolic changes, including increased plasma total TC, LDL-C, FBG, and FBIP < 0.001 , and significantly decreased plasma HDL-C < 0.001 , which supports prior studies (Ayza et al., 2020; Singhal, Singhal, and Singh 2016). However, in the TEL groups, these values were reduced, indicating the clear ameliorative effects of TEL on all metabolic syndrome parameters, showing that TEL can aid in improving the metabolic dysregulation arising from MASLD [14,46]. The common mechanism of action of these metabolic changes is an accumulation of TG related to a compromised balance between TG synthesis, which results in decreased functional insulin signaling [47].

Importantly, TEL treatment in a MASLD model of rats resulted in a significant decrease in hepatic MDA levels ($P < 0.001$), indicating reduced lipid peroxidation and oxidative damage. In addition, TEL treatment resulted in a significant increase in hepatic catalase ($P < 0.001$), GSH ($P < 0.01$), and SOD ($P < 0.001$) levels, indicating the restoration of antioxidant status. This observation is consistent with previous reports showing that TEL has antioxidant properties through stimulation of endogenous antioxidant enzyme activity and a reduction in oxidative stress markers, likely through activation of PPAR- γ and inhibition of NADPH oxidase. [48]. Taken together, TEL reverses oxidative stress in MASLD through reducing lipid peroxidation and promoting hepatic antioxidant defense.

To enhance our understanding of the molecular mechanisms underlying this complex disease, we employed bioinformatics analyses to identify and elucidate key potential regulatory pathways involved

in disease progression. We focused on protein-coding genes only when investigating the important biological mechanisms underlying MASLD. KEGG and Reactome enrichment analyses identified biological processes of interest, particularly those related to lipid metabolism disruption, insulin sensitivity, and inflammation regulation. The KEGG pathway enrichment results highlighted MASLD, the AMPK signaling pathway, and the PPAR signaling pathway, which represent highly specific biological activities. The presence of many of the mentioned genes suggests a strong association between inflammatory cytokines and metabolic dysfunction, including IL-6, TNF, Cpt1a, PPARG, Srebf1, and FASN. The inclusion of AMPK and PPARGs highlights the relevance of energy regulation balance and lipid oxidation function, as they are related to disease [49], and how chronic inflammation induced by inflammatory cytokines, such as IL6 and TNF, frequently complicates MASLD in humans [50–52]. Therefore, inflammation and metabolism interact with one another in a distinct and important workload in MASLD associated with obesity, and targeting this interaction may have therapeutic implications.

Key genes, such as IL6, TNF, Cpt1a, Ppara, Srebf1, and Fasn, were consistently identified among the differentially expressed genes in MASLD, establishing a connection between inflammatory cytokines and metabolic dysfunction, as well as lipid regulation [49,53]. PPAR signaling is associated with genes involved in lipid metabolism and inflammation, with a protective effect on both PPAR α and PPAR γ , which increases fatty acid oxidation and modulates subsequent inflammatory responses [19]. Driven by IL-6, TNF, and other chronic inflammatory conditions, inflammation exacerbates metabolic dysfunction in MASLD, inducing insulin resistance, increasing lipolysis, and contributing to hepatic fat accumulation. Furthermore, these cytokines lead to oxidative stress and fibrogenesis, thereby driving the progression of MASLD [54,55]. These findings highlight the unique and overlapping roles of inflammation and metabolism in MASLD, as determined by bioinformatics and KEGG pathways, supporting the promise of treating MASLD through these pathways.

In addition to the KEGG findings, the reactome analysis confirmed roles in lipid metabolic pathways related to the selected genes and pathways, particularly metabolism of lipids, fatty acid metabolism, and triglyceride metabolism, which included enriched genes such as Cpt1a, PPARG, FASN, ACACA, and Fabp1. Although the bioinformatics enrichment analysis did not explicitly identify the Nrf2/HO-1 axis, the present data and expected contributions in the literature are strongly aligned to show that the Nrf2/HO-1 axis is fundamental to liver protection, as Nrf2 drives the expression of antioxidant defenses, such as HO-1 (and together, they achieve redox balance and suppress proinflammatory mediators, such as TNF- α and IL-6, which are involved in the progression of MASLD [17]. Similarly, MMP-9 and TIMP-1 are not key drivers of enrichment; however, they are well documented in the literature as key regulators of extracellular matrix turnover. Specifically, MMP-9 is associated with degradation of the extracellular matrix, while TIMP-1 inhibits MMP activity; their imbalance is a key feature of fibrotic progression [56,57]. The highly advocated roles of these genes in the associated manuscripts indicate that they have functional and clinical importance in MASLD. These evidences reinforce their potential for diagnostic and therapeutic utility, even when they are not prominently featured in the computational pathway outputs.

Although the occurrence of some classic inflammatory genes, such as IL-6 and TNF, was observed, other genes were not among our resultant genes. On the other hand, many genes identified, including PPARG, FOXO1, and SREBF1, are key regulators of metabolic homeostasis and are known to cross-regulate inflammatory responses through crosstalk with cytokine and stress signaling pathways. This suggests a potentially indirect role in modulating inflammatory cascades in the liver microenvironment. Therefore, we undertook the research of a select number of functionally and empirically relevant genes, as demonstrated in the literature, as well as bioinformatics analysis outputs. For example, we used important genes, such as IL-6, TNF, NF- κ B, Nrf2, HO-1, MMP-9, and TIMP-1, in this study; these genes represent key molecular pathways in hepatocyte inflammation, oxidative stress, and extracellular matrix remodeling and are considered critical to the initiation and progression of MASLD. Many of these genes have also been widely accepted in the literature as biomarkers and therapeutic modulators in the disease state of MASLD through preclinical and clinical studies.

The proinflammatory cytokines TNF- α and IL-6 are key contributors to their synergistic roles in MASLD pathogenesis; they are also critical mediators of hepatic inflammation, metabolic dysregulation, and fibrosis, reflecting the severity and stage of the disease. We demonstrated that, compared to control rats, TNF- α and IL-6 were significantly increased during MASLD development, demonstrating their

involvement in MASLD pathology. TNF- α is a key mediator of the initial and continuous hepatic inflammation phase, specifically activating Kupffer cells and promoting NF- κ B signaling, which leads to oxidative stress, hepatocellular injury, and fibrogenesis. Experimental models have shown that TNF- α inhibition leads to a significant decrease in insulin resistance, liver inflammation, and steatosis [20]. It is also involved in alcohol metabolism by inhibiting alcohol dehydrogenase (ADH), which is associated with elevated endogenous ethanol concentrations in MASLD. Additionally, heightened endogenous ethanol levels can be diminished with TNF- α inhibition, suggesting that exogenous and endogenous ethanol disruptions occur simultaneously as manifestations of flexibility in liver dysfunction [21]. On the other hand, IL-6 is a pro-inflammatory cytokine; it plays a more persistent role in the chronic, low-grade inflammation characteristic of MASLD. It has been proposed as a reliable biomarker of systemic inflammation in metabolic disorders and correlates well with disease severity [23]. Mechanistically, it is capable of disrupting insulin signaling via the upregulation of the suppressor of SOCS-3 and contributes to further insulin resistance and metabolic stress in the hepatocyte [22].

Investigating the role of TEL treatment, the data shown in Figure 6 indicate that MASLD significantly elevated ($P < 0.001$) hepatic TNF- α and IL-6 compared to the control group; however, TEL treatment significantly reduced ($P < 0.001$) these proinflammatory cytokines in MASLD rats. The reduction in inflammatory cytokines may be attributed to TEL inducing PPAR- γ , which in turn inhibits NF- κ B-induced cytokine production, thereby leading to reduced hepatic inflammation. Other PPAR- γ agonists, such as astragaloside IV and pentoxifylline, have similarly exhibited trends in decreasing TNF- α and IL-6 by blocking oxidative stress and NF- κ B signaling [24]. TEL is an ARB with partial PPAR- γ activity, and several studies and meta-analyses have shown that TEL has lower levels of IL-6 and TNF- α compared to standard therapies do. TEL has anti-inflammatory effects by suppressing and modulating NF- κ B activity and gene transcription, which are involved in inflammatory responses [13]. TEL has also been shown to improve lipid metabolism and adipokine balance and, therefore, has the potential to indirectly decrease inflammation, ultimately improving liver histology and disease burden in MASLD [12]. Our findings highlight the important role of TNF- α and IL-6 in the pathogenesis of MASLD, suggesting that the use of agents targeting these cytokines, such as TEL, may provide a potential therapeutic option to reduce inflammation and improve hepatic metabolic activity.

In addition to the direct role of TEL in liver function, there is emerging evidence that TEL may have systemic anti-inflammatory effects by engaging the spleen–liver axis, a central regulatory component of the immune–metabolic interface in MASLD. Angiotensin II receptor blockers (ARBs), such as TEL, differentially inhibit angiotensin II from interacting with AT1 receptors, resulting in decreased vasoconstriction and decreased secretion of aldosterone and inflammatory signaling events [58,59]. A key component of these differences is that mineralocorticoid receptors (MRs) are present in both splenic T lymphocytes and B lymphocytes, modulating the activation and proliferation of immune cells. MR activation leads to a proinflammatory response, while MR antagonism has been found to decrease cytokine production and immune cell infiltration [60,61]. Thus, although there are additional modes of action, TEL-mediated inhibition of the renin–angiotensin–aldosterone system may also indirectly restore splenic immune homeostasis and reduce systemic low-grade inflammation associated with MASLD.

New findings have validated functional signaling biology across the spleen and liver (the spleen–liver axis), which has significant implications for the development of metabolic and fatty liver disorders, including MASLD. Fatty liver inflammation is typified by increased enrichment of myeloid-derived suppressor cells (MDSCs) and natural killer T (NKT) cells in the spleen. Rather than the loss of T and B lymphocytes in the liver compartment, there is stable expression in the spleen compartment. In addition, correlation analysis indicated that the splenic and hepatic distributions of MDSC/NKT cells are well-associated, supporting the concept that the spleen serves as an immune reservoir responsible for modulating liver inflammation resulting from obesity and MASLD [62]. Moreover, mechanistic studies have revealed that leukotriene B4 (LTB4) from the spleen serves as an endocrine signal that enhances TNF production in the liver as a consequence of systemic inflammation, establishing a direct splenic pathway that propagates hepatic injury [63]. Thus, this confirms the current model, in which spleen-mediated signaling contributes to the worsening of inflammation and injury in the liver associated with metabolic disease [64].

Systemic inflammation related to MASLD involves complex immune regulation outside the liver, with the spleen serving as a central hub through signals from proinflammatory cytokines and leukotrienes. TEL may

play a role in opposition to angiotensin II-mediated pathways, as well as in adjustments to immune cell mineralocorticoid receptor activity, which may lead to a sequence of reduced splenic cytokine release, leukotriene-mediated liver inflammation, and downstream changes resulting from immune dysregulation. These findings support the reduction in TNF- α , IL-6, and NF- κ B and improvements in hepatic oxidative stress and histopathology reported in MASLD model studies of TEL [12,65,66]. The liver–spleen axis in nonalcoholic fatty liver disease has also gained some endorsement among healthcare consensus experts, reflecting an acknowledgment of the multiorgan nature of inflammation and the role of the spleen as a reservoir and modulator of hepatic immune regulation. Evidence of a spleno-renal anti-inflammatory network extends the exposures-modulating work of ARBs and opens the door to the potential of a broad-based transition for therapeutic considerations of TEL outside of hepatic immune activity and chronic systemic inflammation potential [64,67].

After inducing MASLD, the levels of Nrf2, HO-1, and TIMP-1 were significantly reduced in the liver, while the corresponding MMP-9 was significantly increased when compared with those in the control groups ($P < 0.001$). These changes support increased oxidative stress in MASLD and a shift towards matrix remodeling or fibrosis. Nrf2 and HO-1 are key components of the antioxidant system, and the activation of Nrf2 leads to increased expression of HO-1, which protects hepatocytes from oxidative stress and inflammation. The downregulation of Nrf2 and HO-1 in MASLD indicates that the antioxidant response has been disabled, therefore contributing to the extent of liver injury and fibrosis [68]. TIMP-1 inhibits the activity of matrix metalloproteinases (MMP-9), which degrade components of the extracellular matrix. A decrease in TIMP-1 and an increase in MMP-9 favor matrix degradation, contributing to liver inflammation and fibrotic changes [69,70]. On the other hand, compared with untreated MASLD rats, MASLD rats treated with TEL presented dramatic increases in Nrf2, HO-1, TIMP-1, and MMP-9 ($P < 0.001$). This implies that TEL reverses antioxidant capacity through the upregulation of Nrf2 and HO-1, which reduces oxidative stress and hepatocyte injury and modulates matrix remodeling through increased TIMP-1 and MMP-9, thereby helping to balance extracellular matrix turnover and facilitating the resolution of fibrosis [12,70]. According to our results, TEL exhibits a combined mechanism of action that enhances antioxidant defenses and regulates fibrotic pathways, which, when combined, may protect against the progression of MASLD and irreversible liver damage.

Compared with controls, the MASLD-induced group exhibited a decrease in hepatic mitochondrial respiratory enzyme complexes concurrent with impaired mitochondrial functional capacity and inefficient oxidative phosphorylation ($P < 0.001$). Mitochondrial dysfunction is a key feature of MASLD and contributes to hepatic steatosis, inflammation, or progression to more severe forms of liver injury by hampering energy production and increasing oxidative stress [71,72]. Overall, TEL treatment significantly improved liver mitochondrial respiratory enzyme complex I activity ($P < 0.001$) and improved liver mitochondrial respiratory enzyme complex II and IV activity ($P < 0.05$) compared to untreated MASLD rats, indicating a significant increase in mitochondrial respiratory function as a result of TEL treatment, likely by augmenting mitochondrial biogenesis and increasing electron transport chain efficiency. The observed changes in mitochondrial enzymes may help combat the metabolic and inflammatory disturbances associated with MASLD, thereby maintaining proper hepatocellular energy metabolism and impeding disease progression. Based on our results, MASLD induces the loss of mitochondrial respiratory enzyme activity; however, TEL induced partial beneficial recovery, suggesting that the therapeutic benefit of TEL may be based on improved mitochondrial health and liver function in MASLD.

Histopathological examinations support the biochemical results. Arguably, this study documents an evolution from cell injury to tissue fibrosis in the natural progression of MASLD. We found evidence of tissue fibrosis via H&E staining, Masson's trichrome staining, and immunohistochemical studies of TGF- β in MASLD-TEL-untreated rats. TEL-treated rats presented preservation of hepatic architecture, in addition to the absence of cytoplasmic vacuoles and minimal degenerative features; in contrast, the cellular injury in the untreated MASLD group was drastic and profound. A significant difference in collagen deposition based on Masson's Trichrome staining between the TEL-treated group and the MASLD group. This suggests the TEL-treated group displayed less fibrosis [73,74]. The immunohistochemical findings also revealed that TGF- β expression was substantially reduced in the TEL group, which is consistent with the observed decrease in fibrosis and inflammatory TGF- β expression [75–77]. The mechanism by which TEL exerts antifibrotic effects is likely to occur through the inhibition of the TGF- β and MMP/TIMP pathways.

Conclusion

In conclusion, our integrated methodology, which uses biochemical experimentation, bioinformatics assessment, molecular profiling, and histopathology, provides a comprehensive understanding of TEL's protective activity against MASLD, with the selected genes – IL-6, TNF, NF- κ B, Nrf2, HO-1, MMP-9, and TIMP-1 – representing the fundamental pathological processes of inflammation, oxidative stress, and intolerance. The molecular mediators that align with the canonical antifibrotic mechanism of TEL in modulating TGF- β and the MMP/TIMP ratios validate our molecular assessment, considering the putative therapeutic targets of the agent. The findings of the molecular analysis are consistent with the experimental outcomes, including the ability of TEL to ameliorate key fibrogenic and inflammatory pathways, enhance antioxidant defense, and maintain hepatic architecture. Taken together, these data support the therapeutic potential of TEL in limiting MASLD progression through its targeted modulation of multiple interconnected pathogenic mechanisms.

Author contributions

Yosra Magdy conceived the presented idea. Yosra Magdy, Mohamed Ibrahim Attia, Hanadi A. Alahmadi, Alhamza Younis Hamza, Lamiaa Elabbasy, and Nevien Hendawy developed the theory and performed the computations. Sherif A. Kamar, Tarek Abd El-Galil, Tarek A. Yousef, Abdullah Al-Dakhil, Amal H. Eissa, and Sara Khedr verified the analytical methods. Nevien Hendawy supervised the findings of this work. All the authors discussed the results and contributed to the preparation of the final manuscript.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. Some data may not be made available because of privacy or ethical restrictions.

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